



From Dimeric to Octameric Metallo-Supramolecular Macrocycles Based on Sterically Congested Ligand-assisted Self-Assembly with Zn(II), Cd(II), and Fe(II)

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Two sterically congested 2,2':6',2''-terpyridine-based ligands LA and LB, composed of asymmetrically contiguous terpyridine units, are designed and synthesized for metallo-supramolecular architectures. The significant advantage of this design is that the terpyridines in these ligands have different chemical environments and show a selective coordination ability with each other. For ditopic ligand LA, the self-assembly with Zn(II), Cd(II), and Fe(II) gave the rhombic dimers, which have the same sets of terpyridine signals as ligand LA. The self-assembly behavior of tritopic ligand LB with Cd(II) and Zn(II) are observed for discrete tetramers under thermodynamic control, whereas ligand LB and Fe(II) are assembled to generate a mixture of tetramer, hexamer, and octamer, which are successfully isolated using regular chromatographic separation. Moreover, the sterically congested ligands and metal ions formed stable intermediates to drive the formation of discrete structures, which is also proved by mixing LB and Cd(II) with a precise stoichiometric ratio of 1:1. These supramolecular complexes are thoroughly characterized by NMR spectroscopy, electrospray ionization-mass spectrometry, and 2D traveling-wave ion mobility-mass spectrometry. This conceptually new design of sterically congested ligands provides a novel strategy for precisely controlled supramolecular complexes with diverse architectures.

attentions over the past few decades.^[1–11] For the assemblies of metallo-supramolecules with precise shapes and sizes, of particular importance are the design of the organic ligands and the predictable coordination geometry of transition metal ions. Especially, organic ligands with the diversity and ease of structural modification turned out to be the most effective strategy to drive the development of coordination-driven self-assembly and the relative research fields. In this spirit, numerous unprecedented organic ligands were designed and synthesized, such as multitopic ligands,^[12–14] multiple type ligands,^[15,16] and complementary ligands,^[17] etc.

Among the numerous reports of organic ligands, 2,2':6',2''-terpyridine (tpy) derivatives, which can coordinate with diverse transition metal ions to form a linearly-coordinated pseudo-octahedral <tpy-M²⁺-tpy> connection and ultimate supramolecular architectures, has gained extensive attention in the field of supramolecular self-assembly.^[18–20] Up to the present, a variety of discrete met-

Coordination-driven self-assembly, which is famous for its high predictability and controllability in fabricating diverse supramolecular architectures, has been received considerable

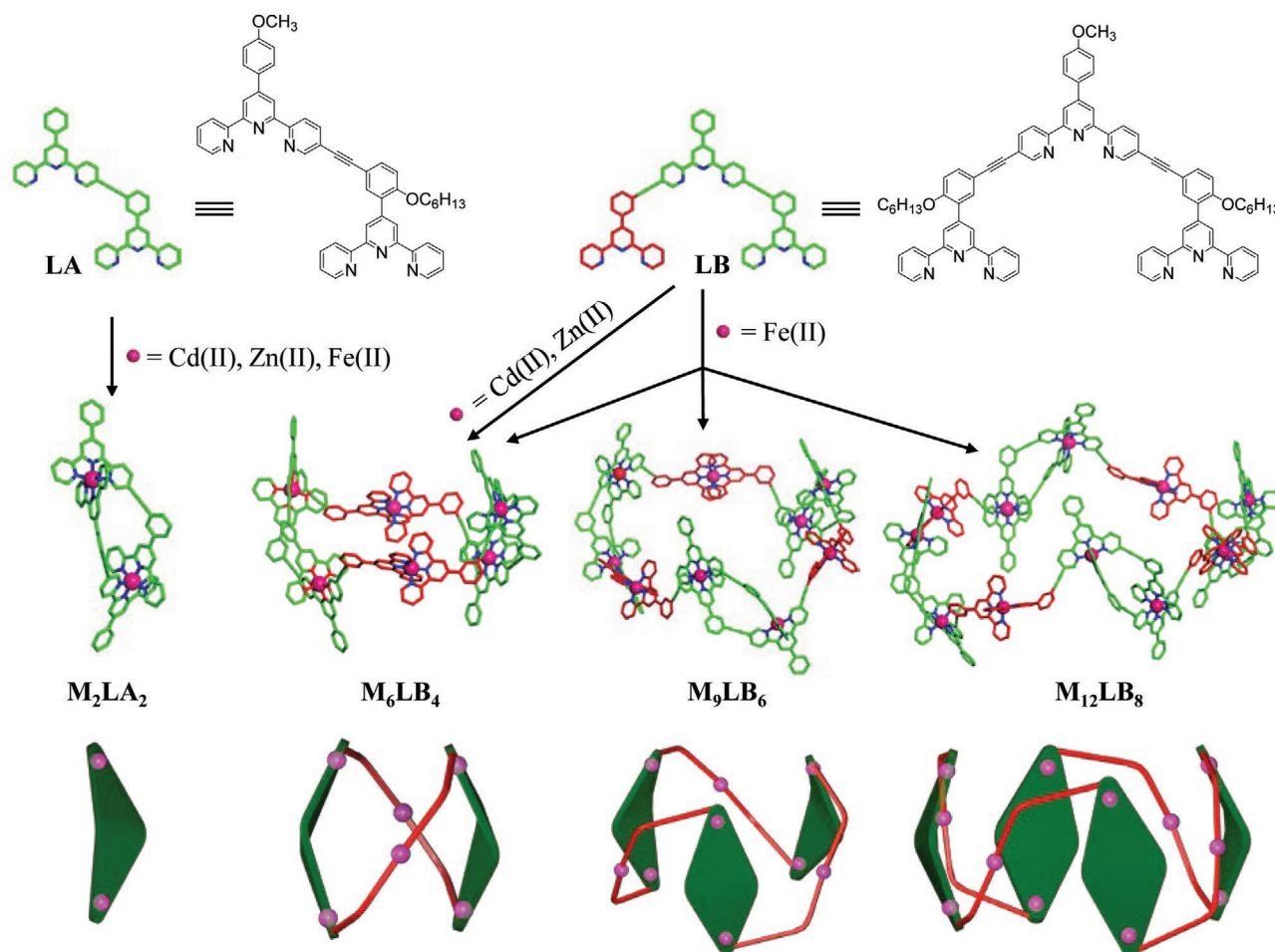
allo-supramolecular architectures based on tpy groups have been extensively reported, including spoked wheels,^[21] pentagram,^[22] hexagons,^[23] cube,^[24] and Sierpiński triangles.^[25] With increasing size and complexity for expected structure, the self-assembly using multi-terpyridinyl monomers as ligand still faces more challenges. Especially, the existence of low coordination selectivity, due to the similar coordination abilities of terpyridines (e.g., binding strength and reversibility) in multicomponent ligands, caused the formation of unpredicted metallo-supramolecules and significantly reduced the controllability of the assembly procedures.^[26] Utilizing bridged terpyridine ligands by stable <tpy-Ru(II)-tpy> connectivity can significantly improve selectivity and form the desired products.^[27,28] However, this led to undesirable difficulty in purification and a low yield of the expected product. Chan et al. reported the complementary terpyridine ligand pairing strategy, which can significantly improve coordination selectivity and lead to the generation of a series of metallo-supramolecular architectures with high complexity.^[17] However, it remains a formidable

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DOI: 10.1002/marc.202000095



Scheme 1. Self-assembly of four complexes using LA and LB. The methoxy and alkyl chain are omitted for clarity.

challenge to design the simple but versatile tpy ligands with high coordination selectivity for diverse architectures and desired controllability.

Herein, two sterically congested ligands constructed based on 2,2':6',2''-terpyridine were designed and utilized for supramolecular self-assembly (simulated structures and simplified cartoon illustrations were shown in **Scheme 1**). The asymmetrically contiguous tpy units in these novel ligands induced the selective coordination between each other. Rhombus-like dimers were obtained by assembling LA and metal namely, Cd(II), Zn(II), and Fe(II) salts. The self-assembly behavior of tritopic ligand LB with Cd(II) and Zn(II) gave discrete supramolecular tetramer complexes. However, when Fe(II) was used as coordination center, a mixture of tetrameric, hexameric, and octameric macrocycles were obtained based on kinetic and thermodynamic balance. Due to the strong connectivity of <tpy-Fe(II)-tpy>, this mixture was successfully separated using regular silica column chromatography. All of the obtained complexes had even numbers of ligands. It suggested the existence of a dimeric intermediate with high selectivity, which was further proved by mixing LB and Cd(II) with a precise stoichiometric ratio of 1:1. This key intermediate with two uncoordinated tpy units was further assembled with additional metal ions to obtain final macrocycles.

Detailed synthetic routes of ditopic tpy ligand LA and tritopic ligand LB with high steric congestion were shown in Scheme S1, Supporting Information. These two ligands were obtained by a Sonogashira coupling reaction in good yield (70% for LA and 68% for LB). The methoxy and alkoxy chains were introduced to increase the solubility of ligands and simplify ¹H NMR spectrum of the aromatic region. Because the asymmetrically contiguous terpyridyl groups in these two ligands have different chemical environments, the ¹H NMR spectrums of LA and LB exhibited two sets of tpy signals (**Figures 1A and 2A**). The full assignments of ¹H NMR based on the 2D COSY spectrum further illustrated the structures of LA (**Figures S1–S4**, Supporting Information) and LB (**Figures S17–S20**, Supporting Information).

Self-assembly behavior of ligand LA with Cd(NO₃)₂·4H₂O was performed with a precise stoichiometric ratio of 1:1 in a mixed solvent of CHCl₃ and MeOH (1:3, v/v) at 50 °C for 8 h, followed by the addition of excess of NH₄PF₆ (to exchange NO₃[−] with PF₆[−]) in methanol to give a white precipitate (Cd₂LA₂) in 90% yield. In the ¹H NMR spectrum, the signals of the complex Cd₂LA₂ became significantly broader compared to that of the ligand LA, indicating the formation of the coordination structure (**Figure 1A**). The same as ligand LA, Cd₂LA₂ also showed two sets of tpy signals, suggesting the highly symmetric structure of Cd₂LA₂. Ph-H^d, Ph-H^d, and Ph-H^e protons were shifted

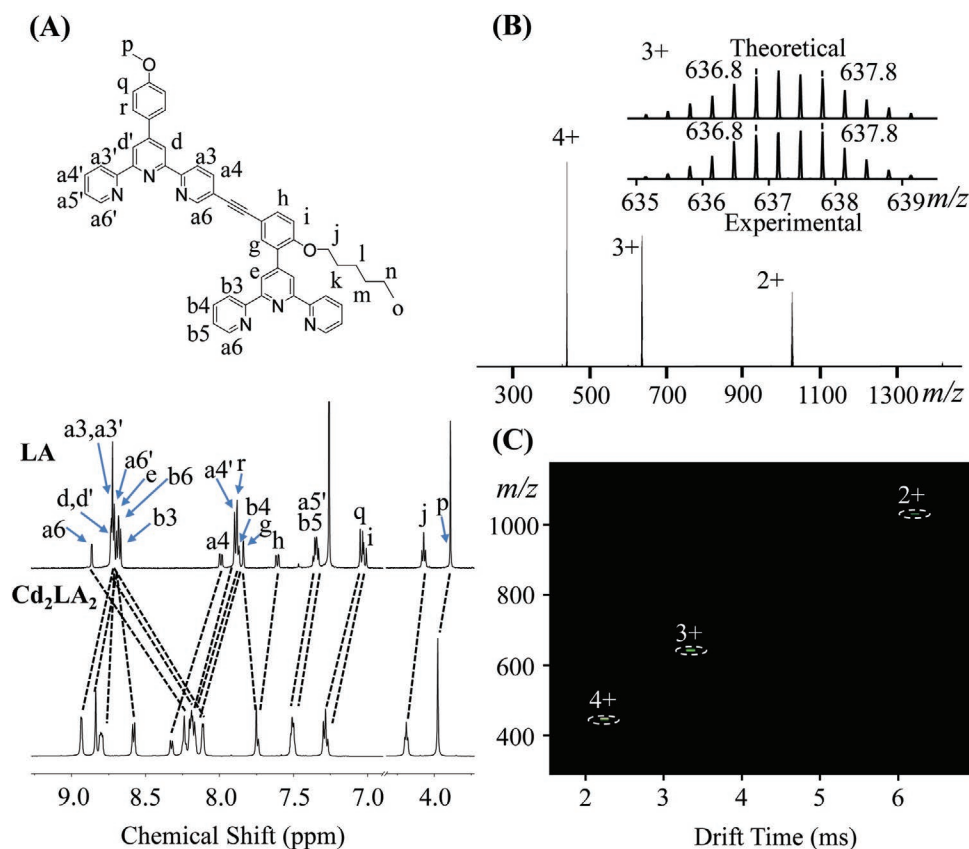


Figure 1. A) ¹H NMR spectra (500 MHz) of ligand LA in CDCl₃ and complex Cd₂LA₂ in CD₃CN. B) ESI-MS and C) 2D TWIM-MS plot (*m/z* vs drift time) of complex Cd₂LA₂. The charge states of the intact assemblies are marked.

to downfield as a result of the lower electron density upon coordination with metal ions. In contrast, due to the electron shielding effect, Ph-H^{a6'}, Ph-H^{a6}, and Ph-H^{b6} were shifted to upfield.^[29] All of the assignments of the proton signals were readily confirmed by the 2D-COSY NMR (Figures S7 and S8, Supporting Information).

The electrospray ionization–mass spectrometry (ESI-MS) results further confirmed the composition of two LA ligands and two Cd²⁺ ions, in which one prominent set of peaks with charge states from 2+ to 4+ were observed (due to the loss of different numbers of PF₆[−]) and the experimental isotope patterns of *m/z* values for each charge state were consistent with the corresponding calculated values (Figure 1B). After the deconvolution, the molecular weight of Cd₂LA₂ was measured as 2346.55 Da, matching well with its expected chemical composition ((C₁₀₂H₈₄O₄N₁₂)Cd₂(PF₆)₄). Traveling-wave ion mobility–mass spectrometry (TWIM-MS) serves as an effective method to determine the molecular composition by providing detailed information for the molecular shapes and sizes.^[30,31] The TWIM-MS spectrum for Cd₂LA₂ showed a series of charge states with narrow drift time distribution ranging from 2+ to 4+, clearly indicating the formation of the expected complex Cd₂LA₂ without any conformer (Figure 1C). By combining the analysis for NMR, ESI-MS, and TWIM-MS detections, it can be confirmed that the two tpy units in ligand LA selectively coordinated with each other to obtain the rhombic dimer architecture as expected. Moreover, we found this dimer is not a

planar structure by simulation due to high steric congestion (Movie S1, Supporting Information).

Complexes Zn₂LA₂ and Fe₂LA₂ were obtained by assembling ligand LA with Zn(II) and Fe(II) (stoichiometric ratio 1:1), respectively. Given that <tpy-Zn(II)-tpy> and <tpy-Cd(II)-tpy> exhibit the similar binding strength,^[32] the identical synthesis procedure as that of Cd₂LA₂ was employed for the self-assembly of LA and Zn(II) to give complex Zn₂LA₂. However, due to the significant discrepancy in the coordination strength and reversibility between Fe(II) and the other two ions, Zn(II) and Cd(II), more demanding assembly condition was required to achieve the desired reversibility of strong connectivity of <tpy-Fe(II)-tpy>. Specifically, the ligand LA and Fe(II) were stirred in a mixed solvent AcOH/CF₃CH₂OH (1:1, v/v) at 125 °C for 24 h. We successfully separated Fe₂LA₂ with a yield of 67% using a regular silica column.^[32,33] Similarly, we characterized the structures of complexes Zn₂LA₂ and Fe₂LA₂ by multinuclear NMR (Figures S9–S12, Supporting Information, for Zn₂LA₂ and Figures S13–S16, Supporting Information, for Fe₂LA₂), ESI-MS, and TWIM-MS (Figures S41 and S42, Supporting Information, for Zn₂LA₂ and Figures S45 and S46, Supporting Information, for Fe₂LA₂). All of the characterizations illustrated that sterically congested ligands assisted self-assembly is an effective strategy to construct supramolecular architectures with higher rigidity and geometric constraints.

Tritopic ligand LB and Cd(NO₃)₂·4H₂O in a stoichiometric ratio (2:3) were mixed in MeOH/CHCl₃ (1:3, v/v) at 50 °C for

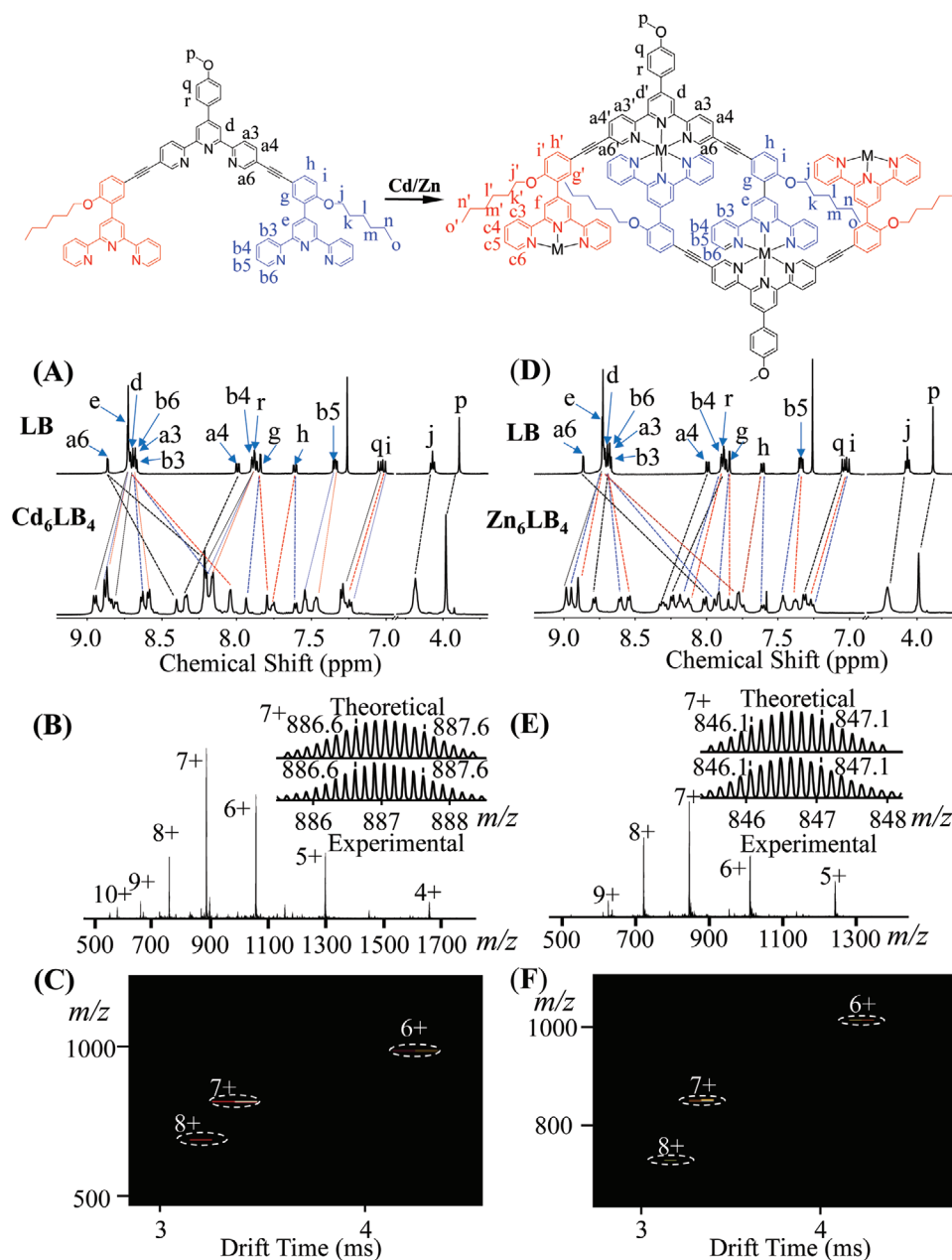


Figure 2. A) ^1H NMR spectra (500 MHz) of ligand LB in CDCl_3 and complex Cd_6LB_4 in CD_3CN . B) ESI-MS and C) 2D TWIM-MS plots (m/z vs drift time) of complex Cd_6LB_4 . D) ^1H NMR spectra (500 MHz) of ligand LB in CDCl_3 and complex Zn_6LB_4 in CD_3CN . E) ESI-MS and F) 2D TWIM-MS plots (m/z vs drift time) of complex Zn_6LB_4 .

8 h, followed by the addition of the excess NH_4PF_6 aqueous solution to obtain a white precipitate (Cd_6LB_4) in a decent yield (90%). The ^1H NMR spectrum of Cd_6LB_4 was shown in Figure 2A. Due to the low tumbling motion of the complex on the NMR time scale, a broad ^1H NMR signals suggested a large complex was observed.^[23,34] For clearly shown assignments, three colors were used for three tpy and contiguous phenyl units. Before complexation, the red tpy and blue tpy are symmetrical and have same signals in ligand LB. After coordination, these two tpy units have different chemical environments and show two sets of tpy signals. In comparison with the ^1H NMR of the ligand LB, Ph-H^{a6} , $\text{Ph-H}^{\text{a6'}}$, Ph-H^{b6} , and Ph-H^{c6}

protons of this complex were significantly shifted to upfield, which indicated that the $\text{Cd}(\text{II})$ assembly was formed. 2D COSY spectrums (Figures S26 and S27, Supporting Information) showed three sets of tpy signals, indicating the formation of a structure with lower molecular symmetry.

ESI-MS spectrum showed intense signals at 758.1, 887.1, 1059.1, 1299.8, and 1660.8 m/z , corresponding to the charge states from 8+ to 4+, respectively (Figure 2B). Each isotope pattern of these peaks was consistent with the corresponding calculated isotope pattern with a molecular weight of 7223.94 Da. This confirmed the formation of the Cd_6LB_4 complex with four LB and six $\text{Cd}(\text{II})$ (Movie S2, Supporting

Information). Further, the TWIM-MS spectrum presented a series of narrowly distributed signals attributed to the continuous charge states of the Cd_6LB_4 , which indicated the unique structure of the assembled complex without any conformer and overlapping isomers (Figure 2C).

We speculated high steric congestion and different chemical environments of the asymmetrical continuous terpyridine units should result in the hierarchical assembling procedures for the ultimate supramolecular complexes. First, the upper tpy unit and one of the nether tpy units in LB (shown in Scheme 1) were preferentially assembled with the ions to form rhombus-like dimer intermediates Cd_2LB_2 (Movie S3, Supporting Information); and then, the intermediates were further assembled using extra Cd(II) and the uncoordinated tpy units to form a more complex Cd_6LB_4 . In order to validate our hypothesis, we analyzed the assembly of LB and $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ at a precise stoichiometric ratio of 1:1. As expected, the obtained molecular weight at 3209.63 Da agreed well with the proposed molecular composition Cd_2LB_2 according to ESI-MS (Figure 3; Figure S52, Supporting Information), suggesting that the dimer intermediates serves as determinative key for the assembly of the ultimate molecular architectures. Further, we thus studied the assembly of ligand LB and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, using the same experimental method as that of Cd_6LB_4 . The obtained tetramer complex Zn_4LB_4 (yield 88%) was carefully characterized by NMR (Figure 2D; Figures S25–S28, Supporting Information), ESI-MS (Figure 2E; Figure S47, Supporting Information), and TWIM-MS analysis (Figure 2F).

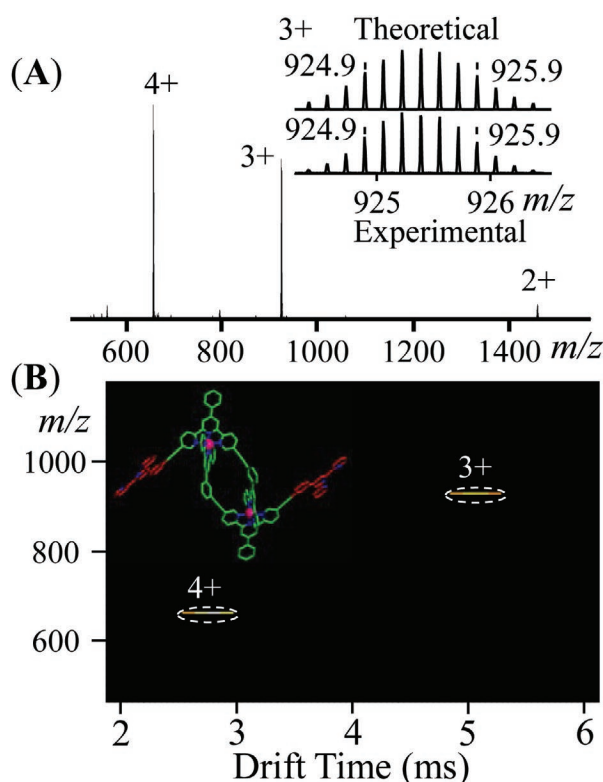


Figure 3. A) ESI-MS and B) 2D TWIM-MS plots (m/z vs drift time) of complex Cd_2LB_2 . The charge states of the intact assemblies are marked.

At 125 °C, LB and $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ with a stoichiometric ratio (2:3) were stirred in a mixed solvent of $\text{AcOH}/\text{CF}_3\text{CH}_2\text{OH}$ (1:1, v/v) for 24 h. In contrast to the unique product obtained using Cd(II) or Zn(II), the self-assembly of tritopic ligand LB and Fe(II) produced three complexes. The Fe_6LB_4 (Movie S2, Supporting Information), Fe_9LB_6 (Movie S4, Supporting Information), and $\text{Fe}_{12}\text{LB}_8$ (Movie S5, Supporting Information) were successfully separated using flash column chromatographed (SiO_2) with isolation yields of 42%, 25%, and 12%, respectively, which were further confirmed by the ^1H NMR spectrum (Figure 4A–C). In addition, the 2D COSY NMR of these three complexes also showed similar three sets of tpy signals as that of Cd_6LB_4 (Figures S32, S36, and S40, Supporting Information). Furthermore, through ESI-MS characterization (Figure 4D,F,H), we identified the architectures of these three assemblies as tetramers (Fe_6LB_4), hexamers (Fe_9LB_6), and octamers ($\text{Fe}_{12}\text{LB}_8$) with molecular weight of 6884.54, 10326.82, and 13769.09 Da, respectively. TWIM-MS indicated the unique structure for each assembly without any conformer or isomers (Figure 4E,G,I). According to these results, all of these three complexes have even numbers of ligands, suggesting dimer structures as building blocks play a pivotal role in the self-assembly.

In summary, we reported the design, synthesis, and characterization of two sterically congested terpyridine-based ligands LA and LB with asymmetrically contiguous terpyridine units. The self-assembly of ditopic tpy ligand LA with Cd(II), Zn(II), and Fe(II) produced rhombus-like dimer structures. The assembly of tritopic tpy ligand LB with Cd(II) and Zn(II) produced tetrameric macrocycles. When Fe(II) was used as the coordination center, a mixture of tetrameric, hexameric, and octameric macrocycles with even number of ligands were obtained and successfully isolated. Preferable dimer assembled by LB and unsaturated Cd(II) (1:1) has proven as an intermediate to induce the formation of the final supramolecular architectures. In view of the potential applications of the coordination-driven supramolecular self-assemblies in host–guest chemistry, chemical separation, drug delivery, and other fields of science and technology, our finding addressed in this work could unlock a new realm of possibilities for fabricating supramolecular self-assembly with high selectivity and predesigned physicochemical properties.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

S.G. and H.Y. contributed equally to this work. This work was supported by the Natural Science Foundation of Jilin Province (20180101297J/C) and Program for JLU Science and Technology Innovative Research Team.

Conflict of Interest

The authors declare no conflict of interest.

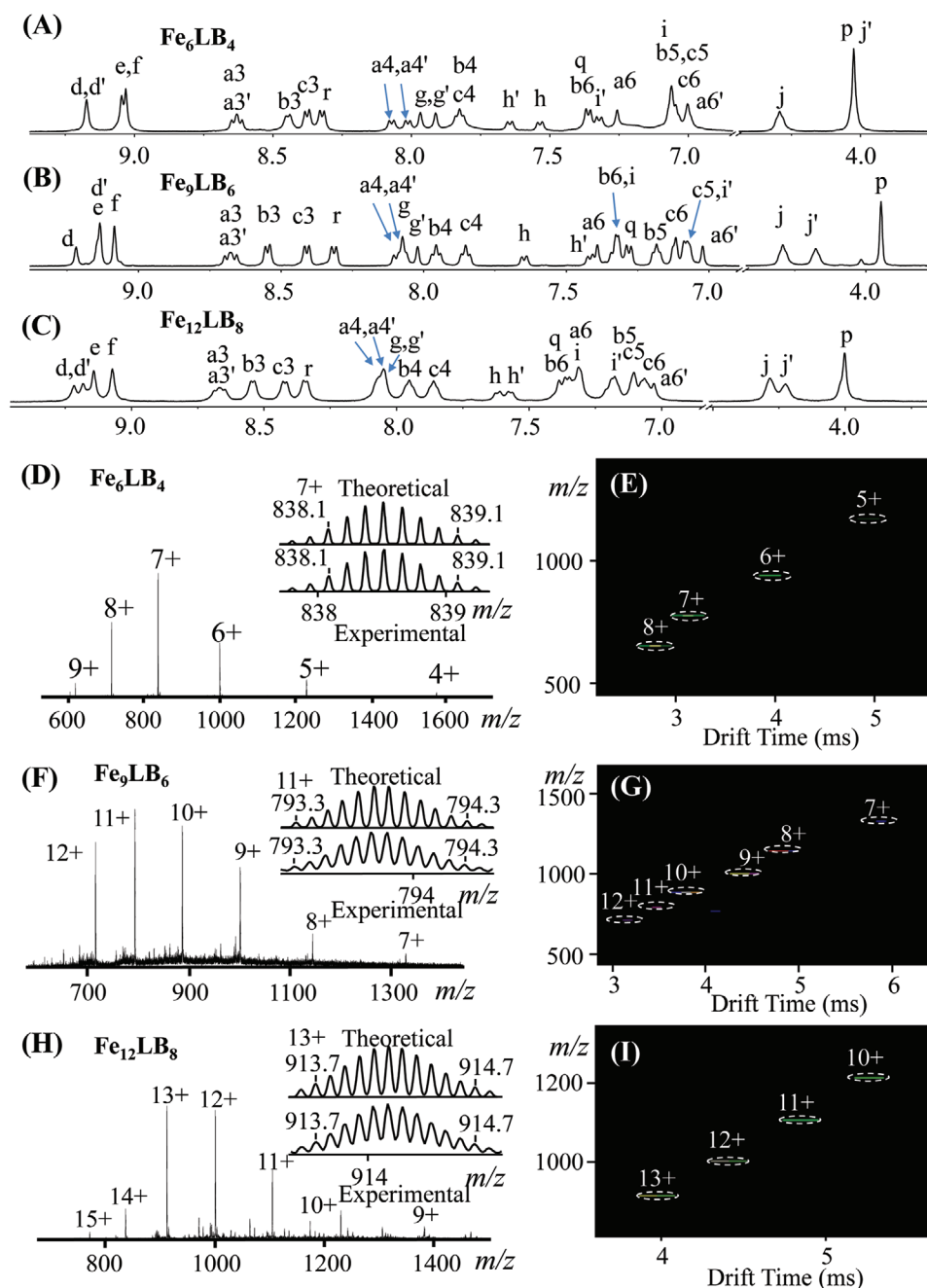


Figure 4. ^1H NMR spectra (500 MHz), ESI-MS spectra, and 2D TWIM-MS plots (m/z vs. drift time) of A,D,E) Fe_6LB_4 ; B,F,G) Fe_9LB_6 ; and C,H,I) $\text{Fe}_{12}\text{LB}_8$.

Keywords

coordination chemistry, self-assembly, terpyridine

Received: February 27, 2020

Revised: March 23, 2020

Published online: April 16, 2020

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